NASSAU CANDY DISTRIBUTOR

Streamlit Dashboard Report

Factory Reallocation & Shipping Optimization Recommendation System

Project Category: Data Analytics, Machine Learning, and Decision Intelligence
Report Purpose: Document dashboard functionality, key performance indicators, analytical findings, model performance, factory-reallocation scenarios, and operational risks.

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1. Executive Overview

The Nassau Candy Factory Reallocation & Shipping Optimization Recommendation System is a data-driven decision-support project designed to analyze historical shipping activity, evaluate predictive models, and explore alternative factory assignments.

The project combines data preparation, exploratory data analysis, machine learning, scenario simulation, and an interactive Streamlit dashboard. It is intended to help business stakeholders understand shipping patterns, examine model results, and evaluate potential improvements in factory allocation and shipping operations.

The dataset contains **10,194 records and 18 original columns** covering orders, shipping modes, customers, geographic information, products, sales, units, gross profit, and cost.

The dashboard provides an interactive environment for exploring the available data and evaluating scenarios. However, the analysis identifies important limitations in the recorded shipping dates and predictive model performance. These issues must be addressed before the system is used for operational decisions.

1.1 Project Objectives

The primary objectives are to:

- Analyze historical order and shipping information.
- Present important business KPIs through an interactive dashboard.
- Examine the distribution of shipping modes.
- Identify potential data-quality problems.
- Compare machine-learning models for shipping lead-time prediction.
- Simulate alternative factory assignments for products and regions.
- Assess operational, financial, and implementation risks.
- Provide recommendations for improving data reliability and decision quality.

2. Dataset Overview

The project uses historical Nassau Candy Distributor data.

Dataset Attribute	Description
Total historical records	10,194
Original columns	18
Order information	Row ID, Order ID, Order Date
Shipping information	Ship Date, Ship Mode
Customer information	Customer ID
Geographic information	Country/Region, City, State/Province, Postal Code, Region
Product information	Division, Product ID, Product Name
Financial information	Sales, Units, Gross Profit, Cost
Engineered information	Shipping lead time and additional analytical features

The feature-engineering workflow extends the original dataset with derived fields for analysis and modelling.

A corrected engineered dataset was also prepared to address date-parsing issues. The dashboard should use the corrected engineered file when its required columns and data types have been validated.

Recommended data source:
`data/Nassau_Candy_Feature_Engineered_Step3_CORRECTED.csv`

The original dataset should be retained for comparison and data-lineage purposes.

3. Dashboard Overview

The Streamlit dashboard is intended to provide an accessible interface for business users to explore the dataset and evaluate potential operational scenarios.

3.1 Key Dashboard Components

A. KPI Summary

The KPI section provides a high-level view of the available dataset and business measures.

Potential indicators include:

- Total historical records.
- Total sales.

- Total units.
- Total gross profit.
- Total cost.
- Average shipping lead time.
- Shipping-mode distribution.
- Number of products and regions represented.

These measures help stakeholders understand the scale and characteristics of the historical data.

Important: The dataset totals should be calculated directly from the corrected file loaded by the dashboard. The number of records is known to be 10,194, but the financial totals and average lead time should not be reported as validated business results until they have been independently checked.

B. Shipping-Mode Analysis

This component displays the number and percentage of records associated with each shipping mode.

It helps users identify the most frequently recorded shipping services and determine where more detailed performance analysis may be useful.

C. Factory Optimization Simulator

The simulator allows users to explore alternative factory assignments for products and regions.

The intended analysis compares candidate assignments with the current baseline. Any results should be interpreted as scenarios rather than confirmed recommendations unless the relevant cost, capacity, and operational constraints have been incorporated.

D. What-If Analysis

The What-If component is designed to support exploration of alternative operating conditions.

Depending on the implemented controls, users may compare scenarios based on shipping priorities, factory assignments, or other available parameters.

Scenario outputs must clearly identify the assumptions used and should not be presented as guaranteed savings or improvements.

E. Recommendation Dashboard

This component organizes candidate factory assignments and other analytical findings into a format intended to support business review.

Recommendations should distinguish between:

- Observed historical results.
- Model predictions.
- Simulated alternatives.
- Recommendations validated using real operational constraints.

F. Risk and Impact Panel

The risk panel communicates data limitations, model uncertainty, and operational dependencies.

Its purpose is to prevent users from interpreting unvalidated estimates as confirmed business outcomes.

4. Key Performance Indicators

KPIs provide a concise view of historical business activity.

KPI	Definition	Business Purpose
Historical Records	Number of records included in the analysis	Measures dataset coverage
Total Sales	Sum of recorded sales values	Shows recorded sales volume
Total Units	Sum of recorded units	Indicates recorded product volume
Total Gross Profit	Sum of recorded gross profit	Supports profitability analysis
Total Cost	Sum of recorded cost	Supports cost analysis
Average Lead Time	Mean of validated shipping intervals	Measures average shipping duration
Shipping-Mode Share	Percentage of records for each mode	Shows the distribution of shipping services
Model MAE	Mean absolute prediction error	Measures average absolute prediction error
Model RMSE	Root mean squared prediction error	Emphasizes larger prediction errors
Model R^2	Coefficient of determination	Assesses predictive performance relative to a mean-based reference
Alternative Factory Candidates	Number of simulated alternative assignments	Measures the scope of the scenario analysis

4.1 KPI Interpretation

KPI values should be interpreted in the context of data quality.

For example, an average shipping lead time is only meaningful if the order dates, ship dates, and date-parsing rules are correct.

Likewise, recorded gross profit and cost should be reconciled with the source system before being used to calculate expected savings or profitability changes from factory reallocation.

The dashboard should display appropriate warnings whenever a KPI relies on unverified data.

5. Shipping-Mode Distribution

The available dataset contains four shipping modes.

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